Convex Functions

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Definition and Variants

Definition 1

$f : \mathcal{C} \to \mathbb{R}$ is **convex** if $\mathcal{C}$ is convex and

$$f(\theta \boldsymbol{x} + (1-\theta)\boldsymbol{y}) \leq \theta f(\boldsymbol{x}) + (1-\theta) f(\boldsymbol{y})$$

for all $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{C}$ and $\theta \in [0, 1]$.

- **Strictly convex:** strict inequality for $x \neq y$, $\theta \in (0, 1)$.
- **Strongly convex ($m > 0$):**

$$f(x) - \frac{m}{2}\|x\|_2^2 \text{ is convex.}$$

Examples of Convex Functions

On $\mathbb{R}$:

- e^{ax} $(a \in \mathbb{R})$
- x^p for $p \geq 1$
- $x \log x$ on $\mathbb{R}_+$

On $\mathbb{R}^n$:

- Affine: $a^\top x + b$
- Norms: $\|x\|_p,\ p \geq 1$
- Least squares: $\frac{1}{2}\|Ax - b\|^2$

Matrix-Valued Examples

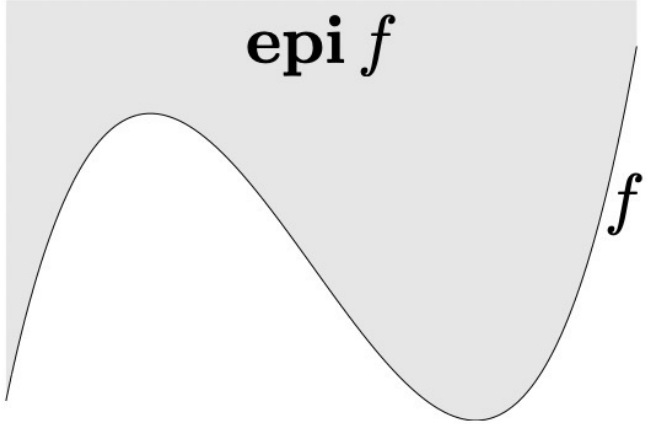
On $\mathbb{R}^{m \times n}$:

- Affine: $\mathrm{tr}(A^\top X)$
- Frobenius norm: $\|X\|_F$
- Spectral norm: $\|X\|_2$
- Log-det (concave): $\log \det X$ on S^n_{++}

Epigraph

epigraph of $f\colon \mathbb{R}^n \to \mathbb{R}$:

$$\mathsf{epi}(f) = \{(x,t) \in \mathbb{R}^{n+1} \mid x \in \mathsf{dom}(f),\ f(x) \le t\}$$



Convexity = "the region above the graph is convex"

Theorem 2

f is convex if and only if $\mathsf{epi}(f)$ *is a convex set*

Jensen's Inequality

Theorem 3

Let $f : C \to \mathbb{R}$ be a convex function where $C \subseteq \mathbb{R}^n$ is a convex set. Then for any $\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_k \in C$ and $\boldsymbol{\lambda} \in \Delta_k$, the following inequality holds:

$$f\left(\sum_{i=1}^{k} \lambda_i \boldsymbol{x}_i\right) \leq \sum_{i=1}^{k} \lambda_i f(\boldsymbol{x}_i).$$

Interpretation: Value at the average $\leq$ average of values.

Note: The domain C must be a convex set to ensure that the convex combination $\sum_{i=1}^{k} \lambda_i \boldsymbol{x}_i$ remains within the domain where f is defined.

Proof

We prove the inequality by induction on k. For $k = 1$, the result is trivial. Assume the hypothesis holds for any k vectors and consider the case for $k+1$ vectors. Let $\boldsymbol{z} = \sum_{i=1}^{k+1} \lambda_i \boldsymbol{x}_i$. If $\lambda_{k+1} = 1$, the result is obvious. If $\lambda_{k+1} < 1$, we observe:

$$\begin{aligned} f(\boldsymbol{z}) &= f\left((1-\lambda_{k+1})\sum_{i=1}^{k}\frac{\lambda_i}{1-\lambda_{k+1}}\boldsymbol{x}_i + \lambda_{k+1}\boldsymbol{x}_{k+1}\right) \\ &\leq (1-\lambda_{k+1}) f\left(\sum_{i=1}^{k}\frac{\lambda_i}{1-\lambda_{k+1}}\boldsymbol{x}_i\right) + \lambda_{k+1} f(\boldsymbol{x}_{k+1}) \end{aligned}$$

Let $\boldsymbol{v} = \sum_{i=1}^{k} \frac{\lambda_i}{1-\lambda_{k+1}}\boldsymbol{x}_i$. Since $\sum_{i=1}^{k} \frac{\lambda_i}{1-\lambda_{k+1}} = 1$, $\boldsymbol{v}$ is a convex combination of k points in C. Applying the induction hypothesis to $f(\boldsymbol{v})$ yields:

$$f(\boldsymbol{z}) \leq (1-\lambda_{k+1})\sum_{i=1}^{k}\frac{\lambda_i}{1-\lambda_{k+1}} f(\boldsymbol{x}_i) + \lambda_{k+1} f(\boldsymbol{x}_{k+1}) = \sum_{i=1}^{k+1}\lambda_i f(\boldsymbol{x}_i) \quad \square$$

Convexity Restricted to a Line

Theorem 4

f is convex on $\mathcal{C}$ iff for every $\boldsymbol{x}$ and direction $\boldsymbol{v}$,

$$g(t) = f(\boldsymbol{x} + t\boldsymbol{v})$$

is convex on its domain.

Restriction of a Convex Function to a Line

example: $f : \mathbb{S}^n \to \mathbb{R}$ with $f(X) = \log\det X$, dom$(f) = \mathbb{S}^n_{++}$

$$\begin{aligned} g(t) = \log\det(X + tV) &= \log\det X + \log\det\left(I + tX^{-1/2}VX^{-1/2}\right) \\ &= \log\det X + \sum_{i=1}^{n} \log(1 + t\lambda_i) \end{aligned}$$

where λ_i are the eigenvalues of $X^{-1/2}VX^{-1/2}$

g is concave in t (for any choice of $X \succ 0$, V); hence f is concave

Key Tool: Spectral Decomposition

To simplify $\log\det(I + tX^{-1/2}VX^{-1/2})$, we use the fact that $M = X^{-1/2}VX^{-1/2}$ is a **symmetric matrix**.

- **Decomposition:** There exists an orthogonal matrix Q ($Q^\top Q = I$) such that:

$$M = Q\Lambda Q^\top, \quad \Lambda = \mathsf{diag}(\lambda_1, \ldots, \lambda_n)$$

  where λ_i are the eigenvalues of $X^{-1/2}VX^{-1/2}$.

- **Identity Transformation:**

$$I + tM = QIQ^\top + tQ\Lambda Q^\top = Q(I + t\Lambda)Q^\top$$

- **Determinant Property:** Since $\det(AB) = \det(A)\det(B)$ and $\det(Q) = 1$:

$$\det(I + tM) = \det(Q)\det(I + t\Lambda)\det(Q^\top) = \det(I + t\Lambda)$$

- **Result:**

$$\det(I+t\Lambda) = \prod_{i=1}^{n}(1+t\lambda_i) \implies \log\det(I+tM) = \sum_{i=1}^{n}\log(1+t\lambda_i)$$

First-Order Characterizations

Theorem 5

Let $f : C \to \mathbb{R}$ be a continuously differentiable function on a convex set $C \subseteq \mathbb{R}^n$. Then f is convex over C if and only if for any $\boldsymbol{x}, \boldsymbol{y} \in C$:

$$f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^\top (\boldsymbol{y} - \boldsymbol{x}) \leq f(\boldsymbol{y})$$

Geometric Meaning

Every tangent plane globally underestimates f.

Proof

- Suppose first that f is convex. Let $\boldsymbol{x}, \boldsymbol{y} \in C$ and $\lambda \in (0, 1]$. If $\boldsymbol{x} = \boldsymbol{y}$, then (1) trivially holds. We will therefore assume that $\boldsymbol{x} \neq \boldsymbol{y}$.
- $\frac{f(\boldsymbol{x}+\lambda(\boldsymbol{y}-\boldsymbol{x}))-f(\boldsymbol{x})}{\lambda} \leq f(\boldsymbol{y}) - f(\boldsymbol{x})$.
- Taking $\lambda \to 0^+$, we obtain

$$f'(\boldsymbol{x}; \boldsymbol{y} - \boldsymbol{x}) \leq f(\boldsymbol{y}) - f(\boldsymbol{x}).$$

- Since f is continuously differentiable, $f'(\boldsymbol{x}; \boldsymbol{y} - \boldsymbol{x}) = \nabla f(\boldsymbol{x})^T (\boldsymbol{y} - \boldsymbol{x})$, and (1) follows.

Proof Contd.

- To prove the reverse direction, assume that the gradient inequality holds.
- Let $\boldsymbol{z}, \boldsymbol{w} \in C$, and let $\lambda \in (0,1)$. We will show that $f(\lambda \boldsymbol{z} + (1-\lambda)\boldsymbol{w}) \leq \lambda f(\boldsymbol{z}) + (1-\lambda) f(\boldsymbol{w})$.
- Let $\boldsymbol{u} = \lambda \boldsymbol{z} + (1-\lambda)\boldsymbol{w} \in C$. Then

$$\boldsymbol{z} - \boldsymbol{u} = \frac{\boldsymbol{u} - (1-\lambda)\boldsymbol{w}}{\lambda} - \boldsymbol{u} = -\frac{1-\lambda}{\lambda}(\boldsymbol{w} - \boldsymbol{u}).$$

- We have

$$f(\boldsymbol{u}) + \nabla f(\boldsymbol{u})^T(\boldsymbol{z} - \boldsymbol{u}) \leq f(\boldsymbol{z}),$$
$$f(\boldsymbol{u}) - \frac{\lambda}{1-\lambda}\nabla f(\boldsymbol{u})^T(\boldsymbol{z} - \boldsymbol{u}) \leq f(\boldsymbol{w}).$$

- Thus,

$$f(\boldsymbol{u}) \leq \lambda f(\boldsymbol{z}) + (1-\lambda) f(\boldsymbol{w}).$$

Second-Order Conditions

Theorem 6 (Second-Order Characterization of Convexity)

Let f be a twice continuously differentiable function over an open convex set $\mathcal{C} \subseteq \mathbb{R}^n$. Then f is convex if and only if $\nabla^2 f(\boldsymbol{x}) \succeq 0$ for any $\boldsymbol{x} \in \mathcal{C}$.

Equivalent condition:

$$v^\top \nabla^2 f(x) v \geq 0 \quad \forall v$$

Proof

Suppose that $\nabla^2 f(\boldsymbol{x}) \succeq \mathbf{0}\ \forall \boldsymbol{x} \in C$. Let $\boldsymbol{x}, \boldsymbol{y} \in C$, then $\exists \boldsymbol{z} \in [\boldsymbol{x}, \boldsymbol{y}] \in C$:

$$f(\boldsymbol{y}) = f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^T(\boldsymbol{y} - \boldsymbol{x}) + \frac{1}{2}(\boldsymbol{y} - \boldsymbol{x})^T \nabla^2 f(\boldsymbol{z})(\boldsymbol{y} - \boldsymbol{x}).$$

$(\boldsymbol{y} - \boldsymbol{x})^T \nabla^2 f(\boldsymbol{z})(\boldsymbol{y} - \boldsymbol{x}) \geq 0 \Rightarrow f(\boldsymbol{y}) \geq f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^T(\boldsymbol{y} - \boldsymbol{x}) \Rightarrow f$ convex.

Proof

Suppose that f is convex over C. Let $\boldsymbol{x} \in C$ and let $\boldsymbol{y} \in \mathbb{R}^n$.
C is open $\Rightarrow \exists \varepsilon > 0$ such that $\boldsymbol{x} + \lambda \boldsymbol{y} \in C \ \forall \lambda \in (0, \varepsilon)$.

$$f(\boldsymbol{x} + \lambda \boldsymbol{y}) \geq f(\boldsymbol{x}) + \lambda \nabla f(\boldsymbol{x})^T \boldsymbol{y}.$$

$f(\boldsymbol{x} + \lambda \boldsymbol{y}) = f(\boldsymbol{x}) + \lambda \nabla f(\boldsymbol{x})^T \boldsymbol{y} + \frac{\lambda^2}{2} \boldsymbol{y}^T \nabla^2 f(\boldsymbol{x}) \boldsymbol{y} + o(\lambda^2 \|\boldsymbol{y}\|^2)$.
Thus, $\frac{\lambda^2}{2} \boldsymbol{y}^T \nabla^2 f(\boldsymbol{x}) \boldsymbol{y} + o(\lambda^2 \|\boldsymbol{y}\|^2) \geq 0$ for any $\lambda \in (0, \varepsilon)$.
Dividing by λ^2, $\frac{1}{2} \boldsymbol{y}^T \nabla^2 f(\boldsymbol{x}) \boldsymbol{y} + \frac{o(\lambda^2 \|\boldsymbol{y}\|^2)}{\lambda^2} \geq 0$.
Taking $\lambda \to 0^+$, we have $\boldsymbol{y}^T \nabla^2 f(\boldsymbol{x}) \boldsymbol{y} \geq 0 \ \forall \boldsymbol{y} \in \mathbb{R}^n$.
Hence $\nabla^2 f(\boldsymbol{x}) \succeq \boldsymbol{0}$ for any $\boldsymbol{x} \in C$.

Log-Sum-Exp

log-sum-exp: $f(x) = \log\left(\sum_{k=1}^{n} \exp(x_k)\right)$ is convex

$$\nabla^2 f(x) = \frac{1}{1^\top z}\mathsf{diag}(z) - \frac{1}{(1^\top z)^2} z z^\top \qquad (z_k = \exp(x_k))$$

to show $\nabla^2 f(x) \succeq 0$, we must verify that $v^\top \nabla^2 f(x) v \geq 0$ for all v:

$$v^\top \nabla^2 f(x) v = \frac{\left(\sum_k z_k v_k^2\right)\left(\sum_k z_k\right) - \left(\sum_k v_k z_k\right)^2}{\left(\sum_k z_k\right)^2} \geq 0$$

since $\left(\sum_k v_k z_k\right)^2 \leq \left(\sum_k z_k v_k^2\right)\left(\sum_k z_k\right)$ (Cauchy-Schwarz inequality)

Convexity-Preserving Operations

- **Nonnegative sum:** $\sum a_i f_i$
- **Affine composition:** $f(Ax + b)$
- **Pointwise maximum:** $\sup_i f_i(x)$
- **Partial minimization:** $\inf_y f(x, y)$

Nonnegative Combinations and Affine Composition

- **Nonnegative combination:** If f_i are convex on $\mathcal{C}$ and $\alpha_i \geq 0$, then
$$f(\boldsymbol{x}) = \sum_{i=1}^{m} \alpha_i f_i(\boldsymbol{x})$$
is convex on $\mathcal{C}$. (Also holds for integrals / pointwise limits.)

- **Affine composition:** If f is convex on $\mathcal{D}$, then
$$g(\boldsymbol{x}) := f(\boldsymbol{A}\boldsymbol{x} + \boldsymbol{b})$$
is convex on $\{\boldsymbol{x} : \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b} \in \mathcal{D}\}$.

Examples:

- **Log barrier (linear inequalities):**
$$f(\boldsymbol{x}) = -\sum_{i=1}^{m} \log(b_i - \boldsymbol{a}_i^\top \boldsymbol{x}), \qquad \mathrm{dom}(f) = \{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{a}_i^\top \boldsymbol{x} < b_i,\ \forall i\}.$$

- **Norm of an affine map:** $\boldsymbol{x} \mapsto \|\boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}\|$ is convex for any norm $\|\cdot\|$.

Pointwise Maximum and Supremum

Rule: If f_s is convex on $\mathcal{C}$ for every $s \in S$, then

$$f(\boldsymbol{x}) := \sup_{s \in S} f_s(\boldsymbol{x})$$

is convex on $\mathcal{C}$.

Examples:

- **Piecewise-linear:** $f(\boldsymbol{x}) = \max_{i=1,\ldots,m} (\boldsymbol{a}_i^\top \boldsymbol{x} + b_i)$.
- **Sum of r largest components:** $f(\boldsymbol{x}) = \boldsymbol{x}_{[1]} + \cdots + \boldsymbol{x}_{[r]}$ (maximum over all $\binom{n}{r}$ choices).
- **Support function of $\mathcal{C}$:** $\sigma_{\mathcal{C}}(\boldsymbol{x}) := \sup_{\boldsymbol{y} \in \mathcal{C}} \boldsymbol{y}^\top \boldsymbol{x}$.
- **Distance to farthest point in $\mathcal{C}$:** $\boldsymbol{x} \mapsto \sup_{\boldsymbol{y} \in \mathcal{C}} \|\boldsymbol{x} - \boldsymbol{y}\|$.
- **Maximum eigenvalue:**

$$\lambda_{\max}(\boldsymbol{X}) = \sup_{\|\boldsymbol{y}\|_2 = 1} \boldsymbol{y}^\top \boldsymbol{X} \boldsymbol{y} \quad (\boldsymbol{X} \in \mathbb{S}^n).$$

Partial Minimization and Distance to Sets

Rule (partial minimization): If $f(\boldsymbol{x}, \boldsymbol{y})$ is jointly convex on $\mathcal{C} \times \mathcal{D}$, then

$$g(\boldsymbol{x}) := \inf_{\boldsymbol{y} \in \mathcal{D}} f(\boldsymbol{x}, \boldsymbol{y})$$

is convex on $\mathcal{C}$.

Examples:

- **Schur complement (quadratic form):** If

  $$f(\boldsymbol{x}, \boldsymbol{y}) = \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} + 2\boldsymbol{x}^\top \boldsymbol{B}\boldsymbol{y} + \boldsymbol{y}^\top \boldsymbol{C}\boldsymbol{y}, \quad \begin{bmatrix} \boldsymbol{A} & \boldsymbol{B} \\ \boldsymbol{B}^\top & \boldsymbol{C} \end{bmatrix} \succeq \boldsymbol{0},\ \boldsymbol{C} \succ \boldsymbol{0},$$

  then

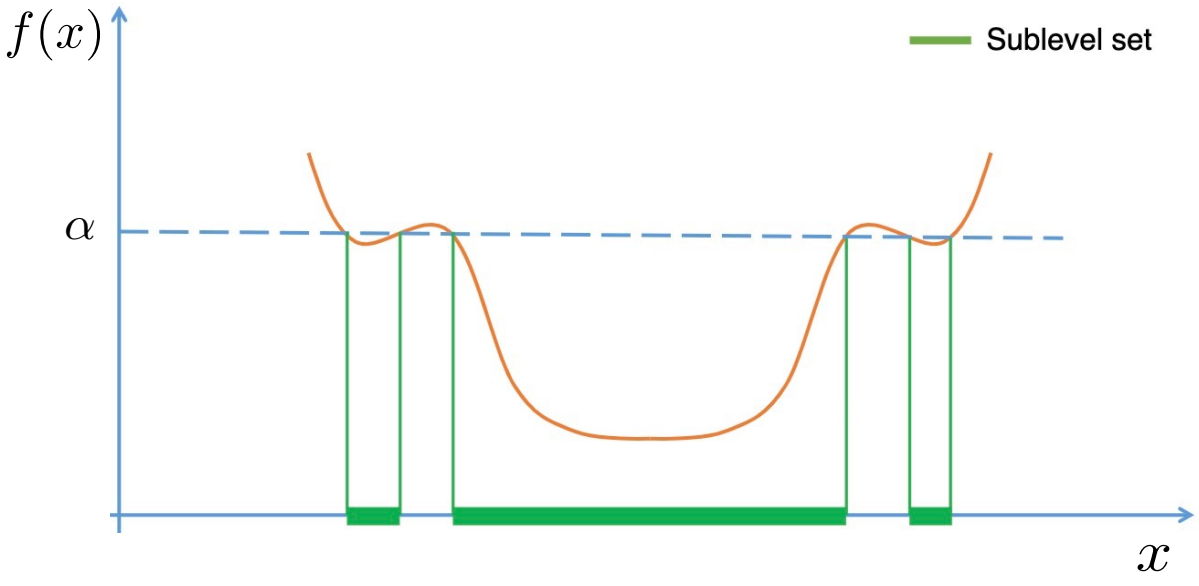
  $$g(\boldsymbol{x}) = \inf_{\boldsymbol{y}} f(\boldsymbol{x}, \boldsymbol{y}) = \boldsymbol{x}^\top (\boldsymbol{A} - \boldsymbol{B}\boldsymbol{C}^{-1}\boldsymbol{B}^\top)\boldsymbol{x}$$

  is convex.

- **Distance to a convex set $\mathcal{C}$:**

  $$d(\boldsymbol{x}, \mathcal{C}) := \inf_{\boldsymbol{y} \in \mathcal{C}} \|\boldsymbol{x} - \boldsymbol{y}\| \quad \text{is convex in } \boldsymbol{x}.$$

Sublevel Set



α-**sublevel set** of $f\colon \mathbb{R}^n \to \mathbb{R}$:

$$C_\alpha = \{x \in \mathsf{dom}(f) \mid f(x) \leq \alpha\}$$

sublevel sets of convex functions are convex (converse is false)

Quasi-Convex Functions

While all convex functions possess convex level sets, the converse is not necessarily true. This leads to a broader class of functions defined by the geometry of their sub-level sets.

Definition 7

A function $f : \mathcal{C} \to \mathbb{R}$ defined over a convex set $\mathcal{C} \subseteq \mathbb{R}^n$ is called **quasi-convex** if for any $\alpha \in \mathbb{R}$, the lower level set (or sub-level set) is convex:

$$\mathsf{Lev}(f, \alpha) = \{\boldsymbol{x} \in C : f(\boldsymbol{x}) \leq \alpha\}$$